AESS Sustainability Hackathon 2026

Participant Submission Example Template

Sample Topic: Smart Study Desk Energy-Saving System

This example is not related to the official competition tracks. It is only used to show participants how a strong submission can be structured, documented, and packaged.

Purpose of this document

Use this document as a formatting and submission-quality reference. It demonstrates how to present the problem, concept, implementation evidence, repository structure, demo video, presentation deck, and final checklist in a clear and judge-friendly way.

Item	Example Entry
Team name	Team Lumos
Example topic	Smart Study Desk Energy-Saving System
Submission type	Training example only
Expected package	Concept document, repository, demo video, presentation deck, and supporting technical files
Important note	Participants must replace all sample content with their own project work.

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1. How to Use This Example Document

This document shows participants what a complete, professional Phase 1 submission package can look like. The example topic is intentionally outside the official competition tracks, so participants focus on structure, clarity, evidence, and packaging rather than copying a technical idea.

Read this carefully

This is not a required topic and not a ready submission. It is a model format. Teams should follow the same discipline: clear problem, focused solution, measurable evidence, accessible files, and a concise explanation.

What participants should learn from this example

- How to write a short concept document that is easy for judges to review.
- How to connect a technical idea to measurable outcomes instead of relying on general claims.
- How to structure a repository so results, code, diagrams, and assumptions are easy to find.
- How to plan a 2-3 minute demo video that proves the project works.
- How to prepare a presentation deck that tells a complete project story.
- How to check the final package before submission.

2. Submission Package Overview

A strong submission package should work as one connected story. The concept document explains the idea, the repository proves the work, the demo video shows the implementation, and the presentation deck communicates the value clearly.

Deliverable	Recommended content	Judge question it should answer
Concept Document PDF	Problem, solution, architecture, evidence, limitations, and impact.	What is the project and why does it matter?
Repository / Technical Files	README, source code, simulation files, results, diagrams, and assumptions.	Can the results be reproduced or checked?
Demo Video	Short walkthrough of the concept, implementation, outputs, and main result.	Can the team demonstrate real work clearly?
Presentation Deck PDF	8-12 slides covering problem, solution, implementation, results, impact, and next steps.	Can the team explain and defend the project in a structured way?
Supporting Files	Datasheets, calculations, screenshots, data tables, diagrams, or additional notes.	Is the evidence credible and transparent?

3. Completed Example Concept Document

Example topic

Smart Study Desk Energy-Saving System - an IoT-based desk lamp and device controller that reduces unnecessary electricity consumption during study sessions. This topic is only a training example and is not part of the official challenge tracks.

3.1 Project Summary

Field	Completed example
Project title	Smart Study Desk Energy-Saving System
Team name	Team Lumos
Problem area	Unnecessary electricity use from desk lamps, chargers, and small study devices left on when students are absent.
Proposed solution	A sensor-based desk controller that detects presence and light level, then switches devices between active, dimmed, and standby modes.
Core proof	Baseline-versus-optimized energy comparison over a simulated 6-hour study period.
Tools used	Arduino-style logic, Python energy model, simple circuit diagram, spreadsheet power budget.

3.2 Problem Statement

Students often leave desk lamps, phone chargers, small fans, and USB devices running during breaks or after they finish studying. Individually, this energy loss is small, but across many students, rooms, and repeated daily behavior, the wasted electricity becomes significant. The problem is not the use of electricity itself; it is the lack of automatic control when devices are no longer needed.

3.3 Proposed Solution

The proposed system is a smart desk controller that uses a passive infrared presence sensor, a light-dependent resistor, and a microcontroller to manage desk-level power consumption. When the user is present, the lamp and connected desk devices operate normally. When the user leaves, the system waits for a short confirmation period, dims the lamp, and then moves selected devices into standby or switches them off based on predefined rules.

3.4 System Architecture

Block	Function	Example implementation
Presence sensing	Detects whether the student is seated at the desk.	PIR sensor or camera-free motion detector
Ambient light sensing	Measures whether the desk already has enough light.	LDR / light sensor
Control unit	Processes rules and decides operating mode.	Arduino Nano / ESP32 / simulation logic
Power switching	Controls lamp and low-power devices.	Relay module or MOSFET switching circuit
Output interface	Shows system mode and energy estimate.	LED indicator or simple dashboard

Architecture flow: Presence sensor + light sensor -> microcontroller logic -> device switching -> energy measurement/output dashboard.

3.5 Operating Logic

1. Read presence and ambient light sensors every 10 seconds.
2. If the user is present and ambient light is low, activate lamp mode.
3. If the user is present and ambient light is sufficient, reduce lamp brightness or keep it off.
4. If no presence is detected for 5 minutes, switch to standby mode.
5. If no presence is detected for 15 minutes, turn off non-essential desk devices.
6. Log estimated energy consumption in each mode.

3.6 Baseline vs Optimized Energy Comparison

The baseline assumes all desk devices stay on continuously during a 6-hour study period. The optimized case assumes the student is away for two 30-minute breaks and that the system applies standby and shutoff rules during absence periods.

Device / Mode	Baseline power	Optimized behavior	Estimated energy over 6 hours
Desk lamp	12 W always on	12 W active, 3 W dimmed, 0 W off	Baseline: 72 Wh / Optimized: 55 Wh
Phone charger	5 W always connected	5 W active, 0.3 W standby	Baseline: 30 Wh / Optimized: 24 Wh
USB fan	4 W always on	4 W active, 0 W off during absence	Baseline: 24 Wh / Optimized: 18 Wh
Controller	0 W in baseline	0.5 W always on	Optimized overhead: 3 Wh
Total	21 W continuous	Adaptive operation	Baseline: 126 Wh / Optimized: 100 Wh

Example result

Estimated energy saving = 26 Wh over 6 hours, equal to approximately 20.6% reduction for this example scenario. The team should clearly state that this is a simulated estimate and explain the assumptions.

Example technologies that may be used

The team may use simple and accessible technologies to build or simulate the solution, such as **Arduino**, **ESP32**, **Raspberry Pi**, **STM32**, **motion sensors**, **light sensors**, **current sensors**, **relay modules**, **smart plugs**, **Python**, **MATLAB**, **Excel**, **Tinkercad**, **Proteus**, or any other suitable simulation or prototyping tool. The selected technologies should be clearly linked to the system function, for example sensing user presence, measuring energy consumption, controlling device operation, collecting data, or visualizing the estimated energy savings.

3.7 Implementation Evidence

- Python model calculates energy consumption for baseline and optimized schedules.
- Controller pseudocode shows the mode-switching logic.
- Circuit diagram shows sensor inputs and switching outputs.
- Results folder includes a comparison chart and a table of assumptions.
- Demo video shows the simulated sensor states and resulting energy calculation.

3.8 Limitations and Future Improvement

- Sensor false positives may occur if the user moves near the desk but does not use it.
- Real energy savings depend on actual device power ratings and student behavior.
- Future versions could use a non-invasive smart plug for measured power instead of estimated values.
- The system should include manual override to avoid user frustration.

3.9 Impact Statement

The project demonstrates a simple but measurable method for reducing wasted electricity in student study environments. Its value comes from clear operating logic, baseline comparison, transparent assumptions, and reproducible calculations. The strongest part of the project is not complexity; it is the clarity of the evidence.

4. Repository Structure Example

A repository should allow reviewers to understand the project without asking the team for missing files. The README should be the starting point and should explain what the project does, how to run the files, and where the results are located.

Folder / File	What it contains in this example
README.md	Project overview, team name, topic, tools, assumptions, how to run the model, and AI usage statement.
/docs	Concept document, architecture diagram, power assumptions, and circuit sketch.
/src	Controller pseudocode or embedded code.
/simulation	Python notebook or script for energy comparison.
/results	Energy comparison chart, summary table, and output screenshots.
/assets	Images used in the presentation deck and demo video.
/hardware	Optional wiring diagram or prototype photos if available.

Example README Skeleton

README.md example structure

Smart Study Desk Energy-Saving System

1. Project overview
2. Problem statement
3. System architecture
4. How to run the simulation
5. Key results
6. Assumptions and limitations
7. AI usage disclosure

5. Demo Video Script Example

The demo video should be short, direct, and evidence-based. A high-quality video is not about cinematic editing; it is about making the technical idea easy to understand in 2-3 minutes.

Time	What to show	Suggested script
0:00-0:20	Title slide + problem image	We are Team Lumos. Our project reduces unnecessary desk electricity use caused by devices left on during study breaks.
0:20-0:45	Architecture diagram	The system uses presence sensing, ambient light sensing, and microcontroller logic to decide when devices should be active, dimmed, on standby, or off.
0:45-1:25	Simulation/code walkthrough	Here is our energy model. We compare a baseline case where all devices stay on with an optimized case using automatic mode switching.
1:25-2:05	Results chart/table	In this example scenario, estimated energy use drops from 126 Wh to 100 Wh over 6 hours, giving about 20.6% savings.
2:05-2:30	Limitations + closing	The result depends on assumptions, so we documented them clearly. Future work would use real smart-plug measurements and improve detection reliability.

6. Presentation Deck Outline Example

Slide	Title	Purpose
1	Project Title and Team	Identify the project clearly.
2	The Problem	Explain the user problem and why it matters.
3	Baseline Scenario	Show what happens without your solution.
4	Proposed Solution	Present the main idea in one diagram.
5	System Architecture	Show inputs, control logic, outputs, and data flow.
6	Implementation Method	Explain code, simulation, prototype, or calculation method.
7	Results	Show baseline-versus-optimized comparison.
8	Impact and Limitations	Explain what the result means and what it does not prove.
9	Repository and Reproducibility	Show where the code, results, and assumptions are located.
10	Final Takeaway	End with a clear conclusion and next step.

7. File Naming and Final Checklist

7.1 File Naming Examples

- TeamLumos_SmartDesk_Phase1_ConceptDocument.pdf
- TeamLumos_SmartDesk_Phase1_Presentation.pdf
- TeamLumos_SmartDesk_Phase1_Video.mp4
- TeamLumos_SmartDesk_Phase1_Code.zip
- TeamLumos_SmartDesk_Phase1_RepositoryLink.txt

7.2 Final Submission Checklist

Check	Item
[ ]	Concept document is clear, concise, and exported as PDF.
[ ]	Repository is public or jury-accessible.
[ ]	README explains how to run the simulation or code.
[ ]	Results are reproducible from the submitted files.

[ ]	Demo video is 2-3 minutes and shows evidence, not only narration.
[ ]	Presentation deck tells a complete story from problem to result.
[ ]	All assumptions, sources, and AI-assisted outputs are disclosed.
[ ]	File names follow the required convention.
[ ]	All links were tested in a private/incognito browser before submission.

Common Mistakes This Template Helps Avoid

Mistake	Better approach
Only describing an idea	Show code, simulation, calculation, prototype, or other implementation evidence.
No baseline comparison	Always compare current/baseline behavior against the improved solution.
Unclear repository	Use README, folders, meaningful file names, and run instructions.
Overclaiming results	State assumptions and limitations honestly.
Weak demo video	Show outputs and explain what the viewer is seeing.
Missing AI disclosure	Mention meaningful AI assistance and confirm the team validated the output.

Appendix A. Blank Participant Template

Instructions

Copy this section into a new document and replace the bracketed text with your own project information. Keep the writing concise, specific, and evidence-based.

A.1 Project Summary

Field	Participant input
Project title	[Insert project title]
Team name	[Insert team name]
Selected topic / track	[Insert selected topic or track]
Problem addressed	[Describe the problem in 2-3 sentences]
Proposed solution	[Describe the solution in 2-3 sentences]
Core proof expected	[What evidence proves the solution works?]
Tools used	[List software, hardware, simulation tools, datasets]

A.2 Concept Document Structure

7. Project summary - one short paragraph.
8. Problem statement - what is wrong with the current situation?
9. Proposed solution - what exactly are you building or simulating?
10. System architecture - show the major blocks and how they interact.
11. Implementation method - explain code, simulation, prototype, calculations, or tools.
12. Results and comparison - show baseline versus improved result.
13. Impact - explain why the result matters.
14. Limitations - be honest about assumptions and weak points.
15. Repository and reproducibility - explain where files are and how to run them.

A.3 Repository Checklist

Folder / File	Status
README.md	[Completed / Not completed]
/docs	[Completed / Not completed]
/src or /simulation	[Completed / Not completed]
/results	[Completed / Not completed]
/assets	[Completed / Not completed]
/hardware if applicable	[Completed / Not completed]

A.4 Demo Video Planning Sheet

Segment	Participant plan
Opening	[Introduce team and problem]
Solution overview	[Show architecture or concept]
Implementation evidence	[Show code, simulation, prototype, or calculation]
Results	[Show comparison chart/table/output]
Closing	[Explain impact, limitation, and final takeaway]

Source Basis Used to Prepare This Example

This template was prepared using the AESH common Phase 1 structure: project repository, concept document, demo video, presentation deck, supporting files, repository expectations, file naming rules, AI usage disclosure, and evaluation logic. The sample project topic itself is fictional and unrelated to the official competition tracks.